

A Counterexample to Conjecture 00000000044

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Claude (Opus 5)[†]

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Abstract

Conjecture 00000000044 asserts that every positive rational is a finite sum of reciprocals of distinct primes, and that a greedy algorithm over prime denominators always terminates. Both fail. If $q = \sum_{p \in S} 1/p$ over a finite set S of distinct primes, then every prime dividing the reduced denominator of q lies in S ; consequently $1/n$ is representable only when $n = 1$ or n is prime, and $1/4$ is the smallest counterexample. The refutation is formalized in Lean 4 against Mathlib.

1 Statement

Conjecture 00000000044 asks two questions and asserts a positive answer to each:

- (i) every positive rational number is a finite sum of reciprocals of distinct primes;
- (ii) the greedy algorithm restricted to prime denominators always terminates in finitely many steps.

Write

$$\mathcal{R} = \left\{ \sum_{p \in S} \frac{1}{p} : S \subseteq \mathbb{P} \text{ finite} \right\}$$

for the set of values representable as in (i), where $\mathbb{P}$ is the set of primes and the elements of S are distinct by virtue of S being a set. For $q \in \mathbb{Q}$ let $\text{den}(q)$ denote the denominator of q in lowest terms.

2 Two obstructions

Proposition 1. *Let $S \subseteq \mathbb{P}$ be finite and $q = \sum_{p \in S} 1/p$. Then*

$$\text{den}(q) \mid \prod_{p \in S} p.$$

Proof. Induction on S . For $S = \emptyset$ the sum is 0, whose denominator is 1. For the step, let $a \in \mathbb{P}$ and $a \notin S$. The denominator of a sum divides the product of the denominators, so

$$\text{den}\left(\frac{1}{a} + \sum_{p \in S} \frac{1}{p}\right) \mid \text{den}\left(\frac{1}{a}\right) \cdot \text{den}\left(\sum_{p \in S} \frac{1}{p}\right) = a \cdot \text{den}\left(\sum_{p \in S} \frac{1}{p}\right),$$

using $\text{den}(1/a) = a$ for a prime. By the inductive hypothesis this divides $a \prod_{p \in S} p$, which is the product over $S \cup \{a\}$. $\square$

Corollary 2. *If $q \in \mathcal{R}$ then $\text{den}(q)$ is squarefree.*

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Proof. A product of distinct primes is squarefree, and a divisor of a squarefree number is squarefree. $\square$

Corollary 2 already refutes (i): $\text{den}(1/4) = 4$ is not squarefree, so $1/4 \notin \mathcal{R}$. The next observation is sharper and shows that the failure is not isolated.

Proposition 3. *Let $S \subseteq \mathbb{P}$ be finite and $q = \sum_{p \in S} 1/p$. Then every prime dividing $\text{den}(q)$ belongs to S . Consequently*

$$q \geq \sum_{\substack{r \text{ prime} \\ r \mid \text{den}(q)}} \frac{1}{r}. \quad (1)$$

Proof. Let r be a prime dividing $\text{den}(q)$. By Proposition 1, $r \mid \prod_{p \in S} p$, so r divides some $p \in S$; as both are prime, $r = p \in S$. Thus the set of prime divisors of $\text{den}(q)$ is a subset of S , and (1) follows because all the omitted terms $1/p$ are positive. $\square$

3 The conjecture is false

Theorem 4. *For $n \geq 2$, if $1/n \in \mathcal{R}$ then n is prime. In particular $1/n \notin \mathcal{R}$ for every composite n , and the smallest counterexample to (i) is*

$$q = \frac{1}{4}.$$

Proof. Let $n \geq 2$ be composite and suppose $1/n \in \mathcal{R}$. We have $\text{den}(1/n) = n$. Let r be the least prime factor of n . Since n is composite, $r < n$. By (1),

$$\frac{1}{n} \geq \sum_{\substack{s \text{ prime} \\ s \mid n}} \frac{1}{s} \geq \frac{1}{r} > \frac{1}{n},$$

a contradiction. (For n prime the same inequality reads $1/n \geq 1/n$ and is no obstruction; indeed $1/n = \sum_{p \in \{n\}} 1/p$.) $\square$

So (i) fails for $1/4$, $1/6$, $1/8$, $1/9$, $1/10$, $1/12, \dots$ — for every composite n . Statement (i) of the conjecture is false.

Corollary 5. *Statement (ii) is false as well.*

Proof. A terminating run of a procedure that repeatedly subtracts reciprocals of distinct primes from a positive rational and stops at 0 exhibits the starting value as a finite sum of reciprocals of distinct primes, that is, as an element of $\mathcal{R}$. Since $1/4 \notin \mathcal{R}$, no such procedure terminates on $1/4$ — whatever rule it uses to choose the next prime, greedy or otherwise. $\square$

Remark 6. The two obstructions are necessary conditions only, and we do not determine $\mathcal{R}$. Corollary 2 and Proposition 3 say nothing about, for instance, whether $1 \in \mathcal{R}$: the denominator 1 is squarefree and has no prime divisors, so (1) is vacuous there. A repaired conjecture would at minimum have to restrict to rationals q with $\text{den}(q)$ squarefree satisfying (1).

4 Formal verification

The accompanying Lean 4 project formalizes the refutation against Mathlib. The conjecture is stated in the formalization, so the final theorem is a formalization of its negation rather than of facts that merely entail it.

- `IsPrimeReciprocalSum, ConjectureHolds` — membership in $\mathcal{R}$, and statement (i).

- `den_dvd_prod` — Proposition 1, by induction on the finite set using Mathlib's `Rat.add_den_dvd`.
- `prod_primes_squarefree, squarefree_den_of_isPrimeReciprocalSum` — Corollary 2, with Squarefree Mathlib's notion.
- `primeFactors_den_subset, sum_primeFactors_den_le` — Proposition 3 and the inequality (1).
- `not_isPrimeReciprocalSum_one_div_composite` — Theorem 4.
- `conjecture_00000000044_false` — $\neg \text{ConjectureHolds}$, by instantiating at $q = 1/4$.
- `no_prime_greedy_terminates_at_one_div_four` — the corollary for (ii), in the rule-independent form stated above.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib v4.33.1. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000044_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.